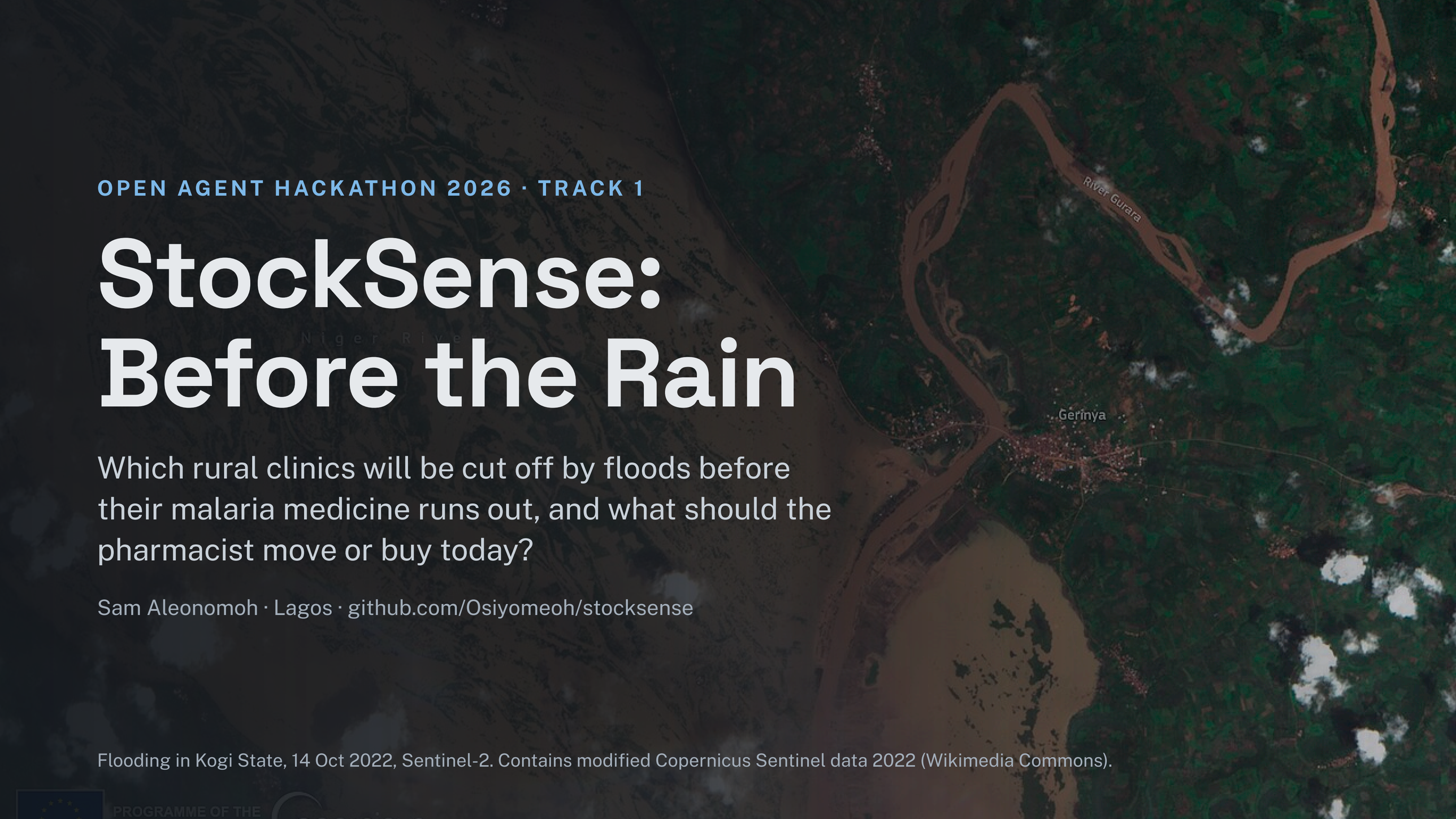
OPEN AGENT HACKATHON 2026 · TRACK 1

StockSense: Before the Rain

Which rural clinics will be cut off by floods before their malaria medicine runs out, and what should the pharmacist move or buy today?

Sam Aleonomoh · Lagos · github.com/Osiyomeoh/stocksense



Flooding in Kogi State, 14 Oct 2022, Sentinel-2. Contains modified Copernicus Sentinel data 2022 (Wikimedia Commons).

PROGRAMME OF THE

THE PROBLEM

In 2022, floods in Kogi State reached 92 health facilities.

66 were partly and 26 completely flooded. Malaria peaks in the same rains. When the road closes before the medicine arrives, the clinic runs out.



A settlement eroded by flooding in Lokoja, Kogi State, Sept 2021. Photo: Sani Maikatanga, CC BY-SA 4.0, cropped.

Source: WHO Africa, 2 Nov 2022: 92 health facilities partly (66) or completely (26) flooded.



Community malaria workers, Kebbi State, June 2015. Badung Samuel Yohanna, CDC / PMI, public domain. Not StockSense users.

WHO IT IS FOR

The LGA pharmacist who keeps a district’s clinics stocked

Their question needs two numbers side by side: how many days the medicine lasts, and how many days until the road floods. No single system holds both.

Not a clinical tool. StockSense drafts a plan; a pharmacist approves every action.

WHOLE-SEASON REPLAY · 4 JUL TO 31 OCT 2022 · REAL RIVER AT LOKOJA

73

clinic-days without medicine
StockSense

93

clinic-days without medicine
naive top-up rule

20 fewer clinic-days, at 26% less spend

₦18,279,200 against ₦24,541,000. Without the river forecast: 128 clinic-days. Without the Critic: 111.

River real (GloFAS); stock, use, roads and supplier lateness SYNTHETIC. Run it: python3 eval/season.py

THE DATA

One Zetaris view joins four sources

Source	Kind	Real or synthetic
Clinic stock and daily use (Postgres on RDS)	database	SYNTHETIC
Road register: the river level that closes each road	spreadsheet	SYNTHETIC
GloFAS river discharge, Lokoja	web API	real, 2022 season
Rainfall, Lokoja (Open-Meteo)	web API	real, 2022 season

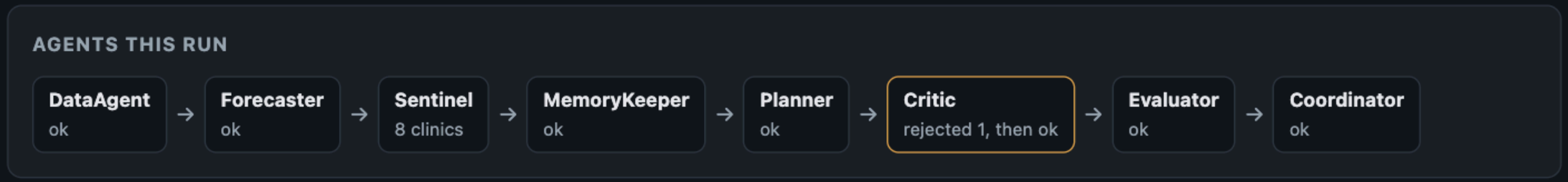
No single source says which clinics will be cut off before their stock runs out. **The join does.**

The view is 38 lines of SQL. The app reads it through 165 lines and never connects to Postgres, the sheet or Open-Meteo itself.

infra/zetaris_usl.sql · Weather data by Open-Meteo.com (CC BY 4.0) · River discharge: GloFAS, Copernicus Emergency Management Service

THE AGENTS

Code decides; the model only words it



Critic

Rejected the first plan: an 8-day purchase to P3, whose road was flagged. The retry used next-day transfers.

Evaluator

Re-queries Zetaris for every number a line rests on. A line it can’t re-verify goes on HOLD.

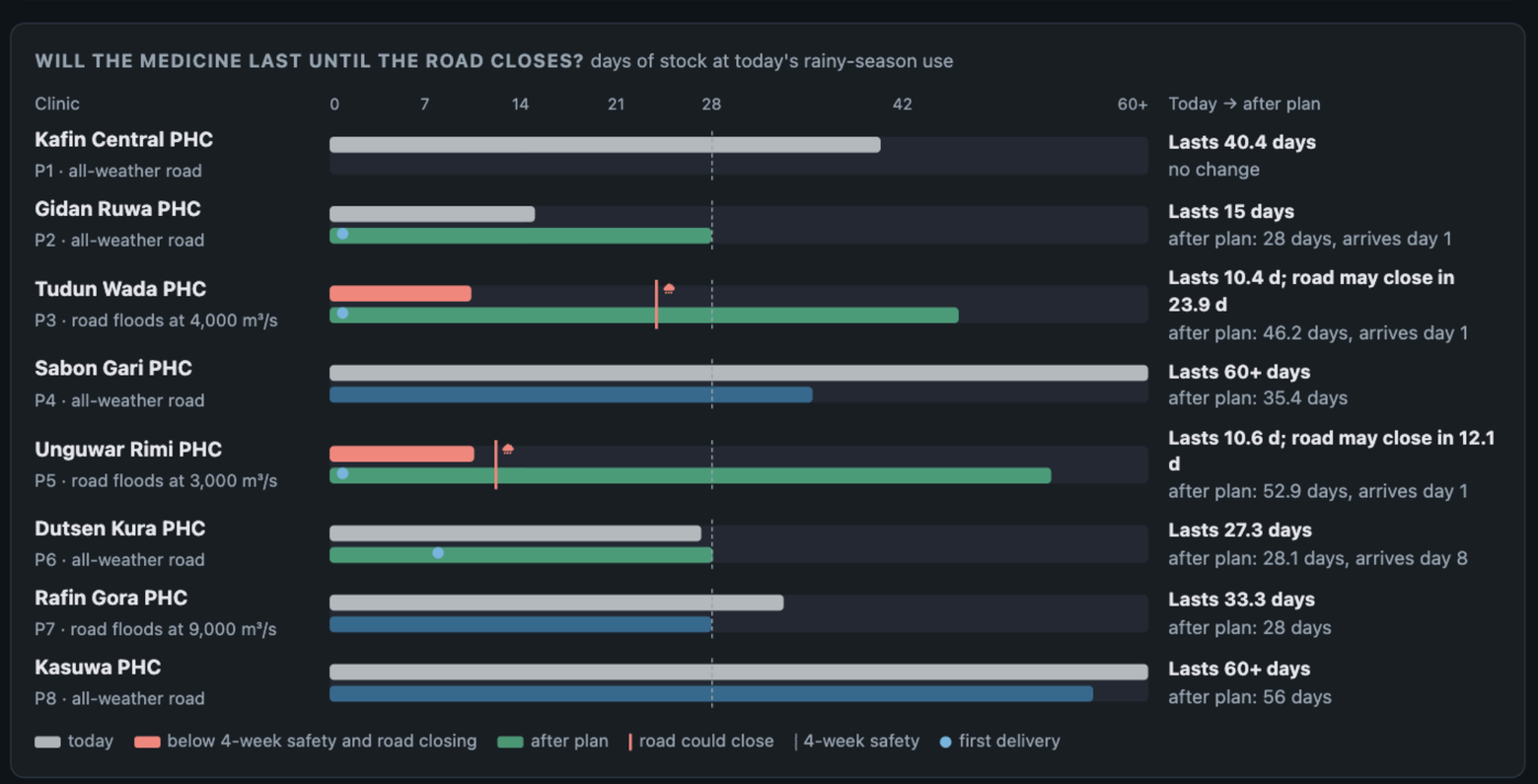
Coordinator

Nemotron, via Nightshift, writes the briefing. A number not in the plan means the fixed template instead.

Every step writes one trace event: logs/trace-<run_id>.jsonl · model nvidia/nemotron-3.5-lightning-30b-a3b

DEMO · SNAPSHOT 2022-08-08, RIVER RISING

Medicine days against road days



10.6 days
of medicine at Unguwar Rimi (P5) today

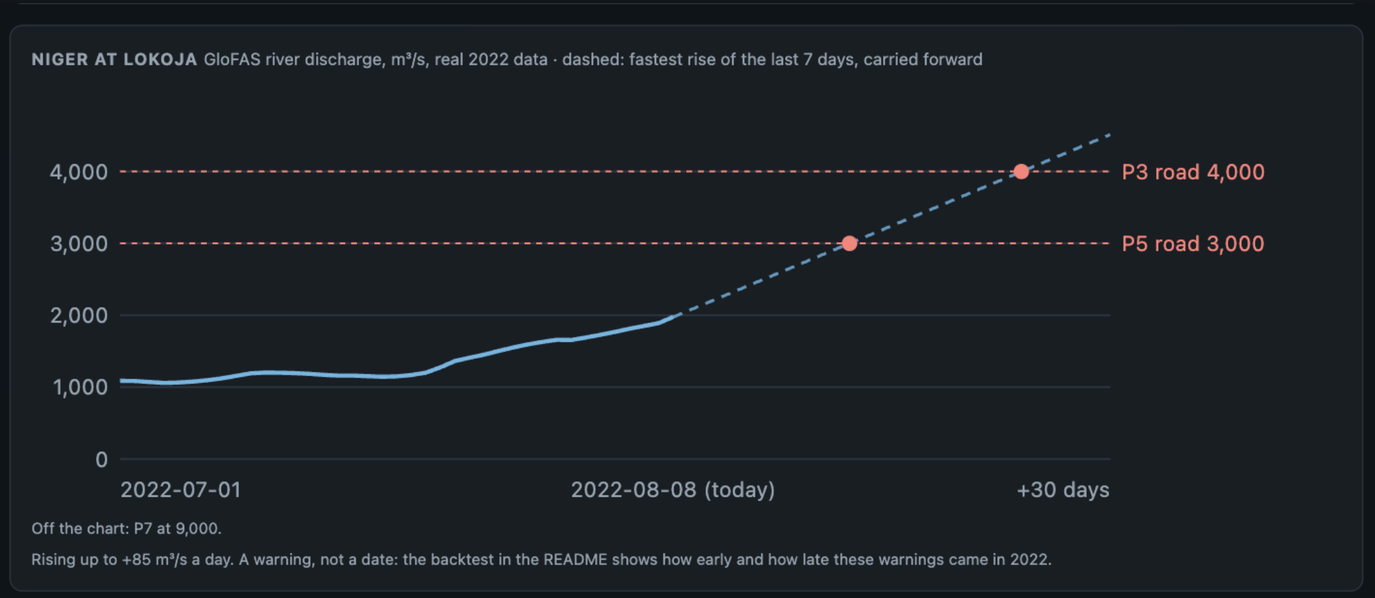
12.1 days
until its road may close

52.9 days
after the plan, delivered on day 1

Plan: ~~₦~~521,900, 7 lines, Critic rejected 1 then ok. Tudun Wada (P3): 10.4 days, road 23.9, after the plan 46.2.

THE FORECAST

A warning, not a date



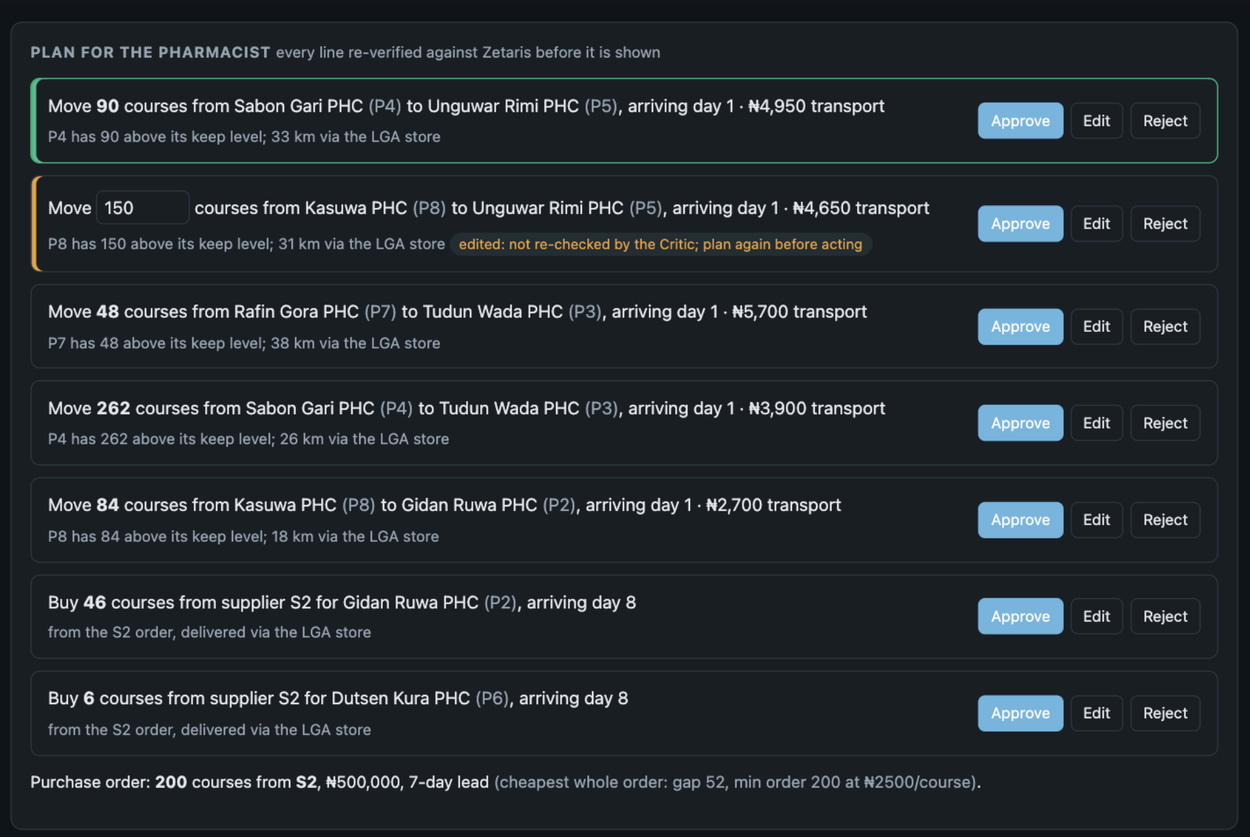
Road	Closed	First warning
P3	28 Aug	30 days ahead
P5	14 Aug	22 days ahead
P7	12 Sep	30 days ahead

Backtested on every day of the real 2022 season, using only readings up to that day. Once a road is flagged, the Critic allows only next-day transfers to it.

python3 eval/backtest_forecaster.py · river real (GloFAS, Lokoja); road thresholds SYNTHETIC

THE HUMAN DECIDES

The pharmacist decides every line



Verified first

Every line was re-checked against Zetaris before it is shown.

Edits are flagged

Approve, edit or reject. An edited line is marked as not re-checked by the Critic.

Nothing is sent

StockSense never places an order or moves stock on its own.

MEMORY · H-MEM BY METERLESS

A note changes the next plan

1 · No notes

200 courses from S2

₦500,000, the cheapest whole order.

2 · “S2 was late again, use S3”

500 courses from S3

Stored at confidence 0.80, so applied. ₦1,000,000, inside the ₦4,000,000 budget.

3 · “S2 is fine now”

0.80 → 0.60: asked, not applied

Below the 0.70 gate, the order goes back to S2 and the pharmacist confirms.



SAFETY

Unsafe input is held or refused

HOLD

2022-07-04 · Dutsen Kura PHC (P6)

The clinic sent no stock count, so it goes on HOLD instead of being guessed. The rest is planned: ~~₦~~514,850, 4 transfers, 1 purchase.

Red team: 10 attacks as tests. 5 found holes, all fixed.

Attack	Before the fix	Now
Model writes a number in words	passed the check	template
Model says the plan “is approved and has been sent”	passed	template
A 130,000-character note	hung	400
Source row with stock −50 or use 0	planned on it, or crashed	HOLD
A bad Content-Length header	crashed the request	400

tests/test_redteam.py: one test per attack. Already safe before: SQL injection in as_of, poisoned and unknown-supplier notes, stock changing mid-run, an over-budget plan.

PROOF

Every number, checked by one command

```
$ python3 prove.py
PASS 07-04 puts P6 on HOLD
PASS 08-08 plan costs ₦521,900
PASS the Critic rejects the 8-day purchase to P3
PASS every number in the briefing is in the plan
PASS after the note: 500 from S3, ₦1,000,000
PASS below 0.70 it is escalated, not applied
PASS backtest row P5 matches the real 2022 river
PASS season: 20 fewer clinic-days, 26% less spend
… and 19 more PASS lines
27/27 claims proven
```

Each claim is checked twice: the README must state it, and a fresh replay must produce it.

No keys and no cluster: it replays the recorded Zetaris and Nightshift responses. 7.7 s on our laptop.

CI runs it with the test suite on every push.

Output of python3 prove.py, 27 Oct 2026; 19 of the 27 PASS lines are left out here to fit.

WHAT WE HIT

What Zetaris should improve

Query	Median	Slowest
clinic_risk, all clinics	5.2 s	46.9 s
clinic_risk, 7 clinics	7.5 s	50.5 s
River up to 2022-08-08	1.4 s	2.4 s
River on 2022-08-08	1.8 s	12.4 s

Zetaris Cloud, 26 Oct 2026, 5 runs per query, from Lagos.

Latency spikes. The same join took 5 s on one run and 47 s on another.

Errors under light load. HTTP 400 TTransportException, with nothing a caller can act on.

REST sources re-fetched on every query, about 3 minutes each. A cache or TTL would remove our copy step.

The SQL editor splits at every “;”, even inside string literals.

On errors the app returns 502 zetaris_unavailable and plans nothing rather than guess.

STACK AND NEXT STEPS

How it is built, and what comes next

BUILT WITH

Eight agents in Python 3, standard library only. Every step writes a trace event.

Zetaris: one USL view, clinic_risk, joins RDS Postgres, a road spreadsheet, the GloFAS river and Open-Meteo rain.

H-MEM (Meterless): supplier notes and road warnings, kept between runs.

NVIDIA Nemotron through Nightshift: the Coordinator's briefing, number-checked, with a template fallback.

Shipping: HTTP API and Docker image. CI runs prove.py and 32 tests on every push.

NEXT

Real data from one LGA: its stock reports and road register, in place of the synthetic clinics and roads.

A pilot with that LGA's pharmacists.

More medicines than malaria treatment.

A stronger river forecast: the August 2022 surge outran the week before.

Approvals by SMS or WhatsApp, where pharmacists already are.

Next steps are plans, not results. Clinics, stock and roads in this demo are SYNTHETIC; river and rain are real 2022 data.

StockSense drafts the plan before the rain. A pharmacist decides.

Code
github.com/Osiyomeoh/stocksense

Recorded runs, no install
osiyomeoh.github.io/stocksense

Check every number
`python3 prove.py`

Not a clinical tool. Clinics, stock and roads are SYNTHETIC; river and rain are real 2022 data. Image: contains modified Copernicus Sentinel data 2022.

PROGRAMME OF THE